[Lecture Note] Sampling

GE2021 Research Technology, Spring 2025

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Fri, 23 May 2025

Learning Objectives

- 6-1 Explain the purpose of sampling.
- 6-2 Compare and contrast probability and nonprobability sampling.
- 6-3 Describe the types of nonprobability sampling.
- 6-4 Summarize the types of probability sampling.
- 6-5 Understand sampling error, confidence interval, and saturation.

SHORT VIDEO INTRODUCTION

https://www.youtube.com/watch?v=pTuj57uXWlk

Sampling

- Sampling: the procedure used by researchers to select a subset of the population that can be used to conduct scientific study.
- Sample: the subset of the population used in sampling.
  - By following sampling procedures, researchers increase the likelihood of getting a representative subset of the population as their sample.
- Population: the entire group of people that are the focus of the study.
- Representative sample: one that reflects the same features of interest as the entire population.
- The first step toward finding the best sampling technique is to determine the population of interest and develop a sampling frame.
  - Sampling frame: a list of the entire population of the study.
  - Researchers have developed different ways to sample from a population even if a sampling frame is not possible.

Probability and Nonprobability Sampling

- Probability: participants are randomly selected and a sampling frame is often used.
- Nonprobability: participants are often not randomly selected and a sampling frame may not be available.
- Randomization for sampling means everyone in the population has an equal chance of participating in the study.
- It is common for qualitative studies to use nonprobability sampling types and for quantitative studies to use probability sampling types.
- Probability sampling often results in a larger sample size while nonprobability sampling results in a smaller sample size.

Types of Nonprobability Sampling

Nonprobability sampling includes various techniques that do not include an equal change of participation. They often involve populations that are difficult to reach or where a sampling frame does not exist.

Convenience Sampling

- Convenience sampling: technique that allows the researcher to select any participants who are available to participate in a study, even if they are not representative of a population.
- Their participation happens by availability and accident, so this is often also called accidental sampling.
- Pilot studies: initial inquiries that pave the way for larger studies.
  - Helps avoid badly conducted larger studies.
  - Cost-effective way to test aspects of a study.

Snowball Sampling

- Snowball sampling: technique in which participants are selected by word of mouth.
  - One or a few participants are willing to help find more participants.
  - May not be representative of the entire population because the researcher selects only participants who know each other.
- A researcher can start with convenience sampling and continue with snowball.
- A researcher should try to introduce as much participant variety as possible by encouraging participants to recommend people with specific characteristics to participate.

Purposive Sampling

- Purposive sampling: technique that allows the researcher to select participants of interest for the study; also called judgmental sampling.
- Homogenous sampling: type of purposive sampling technique in which participants are chosen based on a trait of characteristic of interest.
  - Each participant must have the specific characteristic the researcher is looking for.
- Deviant case sampling: focuses on unusual or very specific cases.
- Purposive sampling allows a researcher to pick participants according to the characteristic under the study, but that means the results of the study cannot be generalized to the greater population and offer low external validity.

Quota Sampling

- Quota sampling: technique that compares different groups within the population of interest; found in both qualitative and quantitative designs.
  - The researcher needs a sample of predetermined quota or proportions.
- Proportional quota sampling: the sample's representation of the same proportion as it exists in the entire population of interest.
- Non-proportional quota sampling: uses a different quota from the one found in the population of interest because the study's aim is to compare two or more different groups of interest.

Types of Probability Sampling

All types of probability sampling use randomization, but they vary in the methodology used to choose participants, which depends on the aim of the study.

Simple Random Sampling

- Simple random sampling: sampling procedure that relies on complete randomization without any specific boundaries.
- Conducted in one step using techniques such as random number generation.
- Every member has an equal chance of participating and it ensures representation of the population.
- It requires a complete list of the population from which the sample is drawn.

Systematic Random Sampling

- Systematic random sampling: same technique as simple random sampling but has a level of organization embedded in it.
- A systematic way of randomly selecting participants from a sampling frame.
- Operates under the assumption that participants are randomly order in a sampling frame so everyone in the list has the same chance of participating in the study.

Stratified Random Sampling

- Stratified random sampling: a technique that become valuable when the study is focused on understanding, comparing, or analyzing different groups of a population.
  - Requires equal numbers of participants from each group.
- Strata: lists of different groups to choose participants from.
- Proportionate stratified sampling: type of sampling that follows the proportions of the population but also creates specific strata that are of interest to the study.
- Disproportionate stratified sampling: a sample in which the proportions are not equivalent to the proportions in the entire population.

Cluster Random Sampling

- Cluster random sampling: useful technique for when the population of interest is widely spread out.
- Researcher follows two steps:
  - Clusters are randomly defined.
  - A cluster is a group of population in a specific geographical area.
  - Participants are randomly selected from each participating cluster.

Sampling Assessment

Sampling Error

- Sampling error is a statistical measurement that broadly represents how different the sample size is from the population of interest.
- Used in quantitative studies.
- Helps to determine whether the sample is sufficiently representative to the entire population.
- A small sampling size indicates a good match between the sample and entire population, which allows researchers to generalize their findings beyond the sample.
- Sampling error is measured for any different parameters of the population that are the focus of the study.

Confidence Interval

- Confidence interval (CI) measures the confidence of lack of confidence in the sample size and sampling method.
- One of the first indexes looked at in a quantitative study.
- Higher sample sizes lead to higher confidence.

Saturation

- Saturation is the measure used to estimate the right sample size in qualitative work.
- Saturation refers to the moment in qualitative data collection where no new information is coming in.
- Reaching saturation means the sample size is sufficient and data collection can stop.

Class Activities:

- You are a member of a research team seeking to collect data from drug users in households across the United States. You need to determine how you will draw this sample. You determine a multistage cluster sampling strategy will be suitable for this project. Draw out your sampling strategy on a board or a sheet of paper.
  - What will be your first stage of sampling, your second, and so on?
  - What are potential limitations and strengths of your sampling strategy?

- Divide the classroom into three groups and ask each group to read the scenario below. Scenario: A researcher is interested in studying gang violence in Chicago and proposes to use a snowball sampling strategy to access gang members to conduct interviews. The researcher has convened a meeting of her team, the university Institutional Review Board (IRB) and community members from urban Chicago to discuss this project. The roles are as follows:
  - Group A: A researcher interested in studying gang violence in Chicago.
  - Group B: Members of a university IRB.
  - Group C: Community members from urban Chicago.
- Each group is trying to advocate for the following:
  - Group A: Your task is to convince Group B and Group C that you have thought through this project and will conduct it safely and ethically.
  - Group B: Your task is to ensure that the researcher is following ethical guidelines and protecting the rights of participants.
  - Group C: Your task is to advocate for the rights of your community.
- Put yourself into your respective role and discuss this project. Your professor will serve as the meeting convener.

- This activity requires students to create a short academic satisfaction survey that could be distributed to other students at their school or university. Ask students to work in small groups and devise the following:
- A draft of a short academic satisfaction survey
- A plan for sampling students to complete this survey. Ask students to write out their plan in - form that would be appropriate for an academic journal. This would include:
  - A description of the population
  - A description of the sample
  - A plan for how the sample will be selected and justification for why this sampling strategy was chosen
  - Limitations of the sampling strategy including a discussion about the generalizability of the findings obtained from this sample

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**
  - A
  - B
- **Format and Requirement**
  - PDF format, < ?? pages
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
  - APA, Double-Space, No title page, 12 points, Reference section if necessary
  - Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)
- due date
  - by Date(DAY) 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference